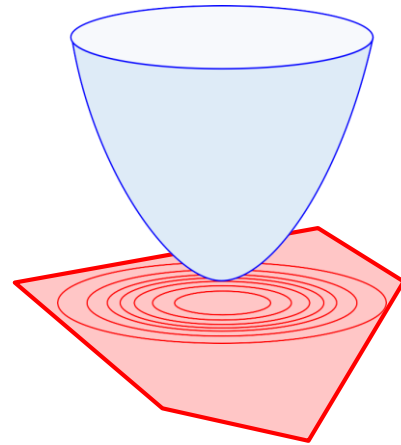




Introduction to Optimization

MATH-224 (OPT): Lecture 1



Tanujit Chakraborty

@ Sorbonne

tanujit.chakraborty@sorbonne.ae



What is Optimization?

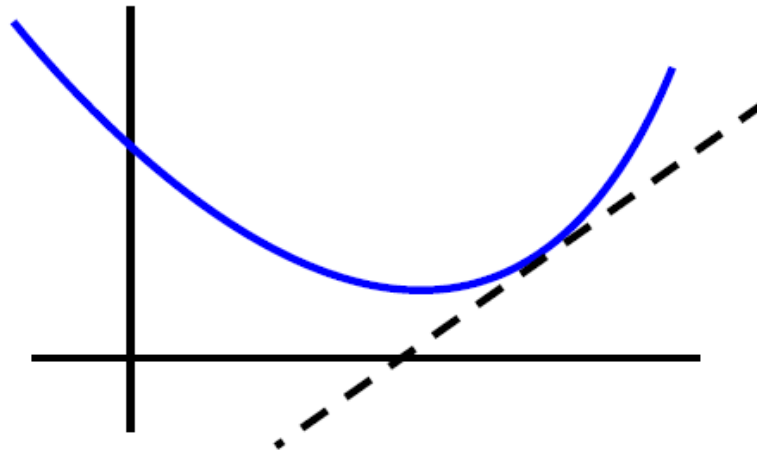
A Cartoon View of Optimization

- Optimization is the mathematical process of finding the **best possible choice** from a collection of feasible choices.
- The function tells us what we want to optimize; the set tells us where we are allowed to look.

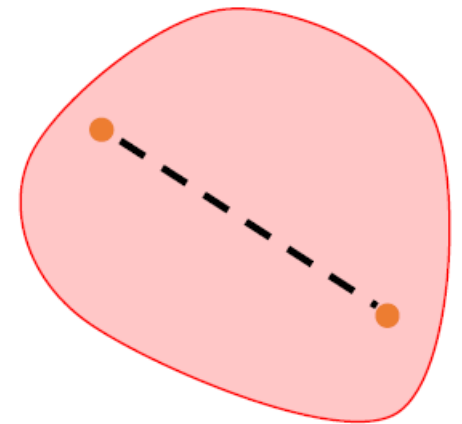
$$\min_{\mathbf{x} \in \mathcal{C}} f(\mathbf{x})$$

$$f : \mathbb{R}^d \rightarrow \mathbb{R} \quad \mathcal{C} \subseteq \mathbb{R}^d$$

Some function (Convex)



Some set (Convex)

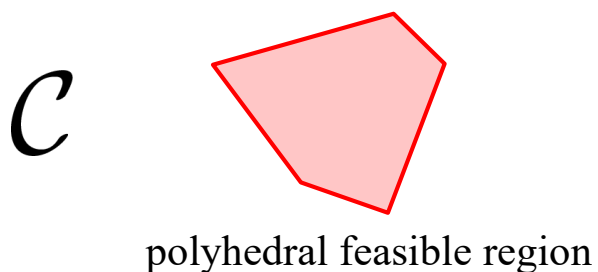
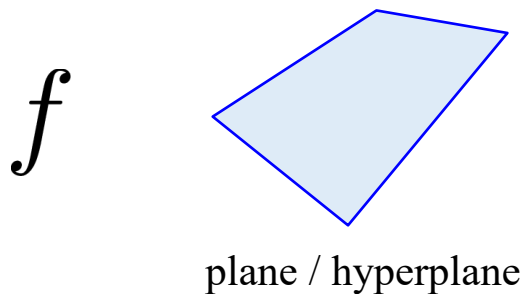


Examples



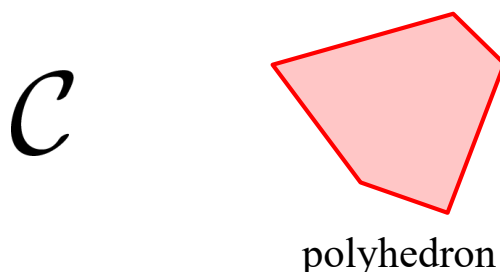
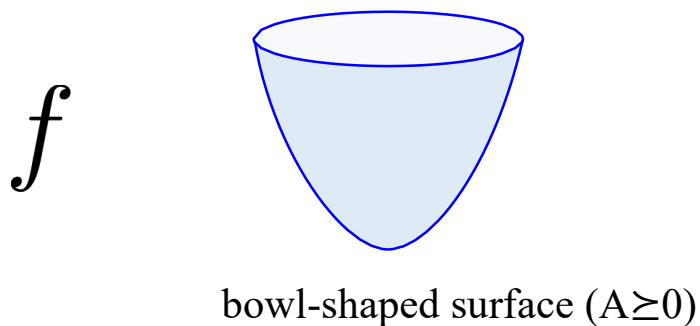
Linear Programming

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^d} \quad & \mathbf{a}^\top \mathbf{x} \\ \text{s.t.} \quad & \mathbf{b}_i^\top \mathbf{x} \leq c_i \end{aligned}$$



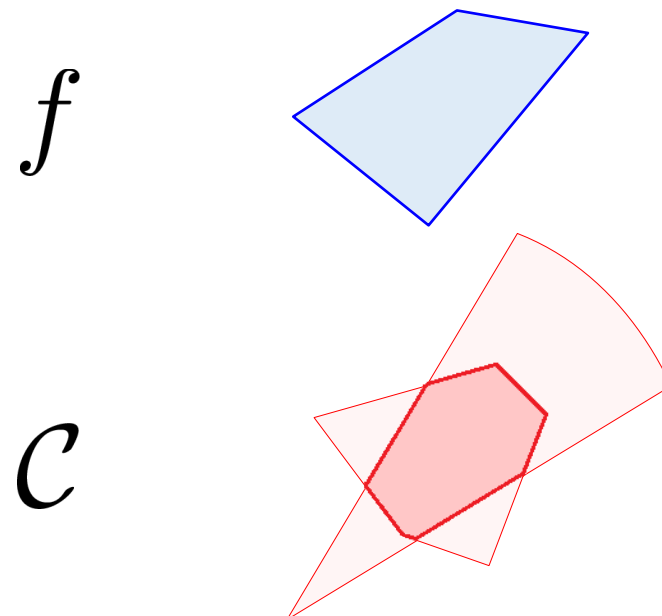
Quadratic Programming

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^d} \quad & \frac{1}{2} \mathbf{x}^\top \mathbf{A} \mathbf{x} + \mathbf{a}^\top \mathbf{x} \\ \text{s.t.} \quad & \mathbf{b}_i^\top \mathbf{x} \leq c_i \end{aligned}$$



Semidefinite Programming

$$\begin{aligned} \min_{\mathbf{X} \succeq 0} \quad & \mathbf{A}^\top \mathbf{X} \\ \text{s.t.} \quad & \mathbf{B}_i^\top \mathbf{X} \leq c_i \end{aligned}$$



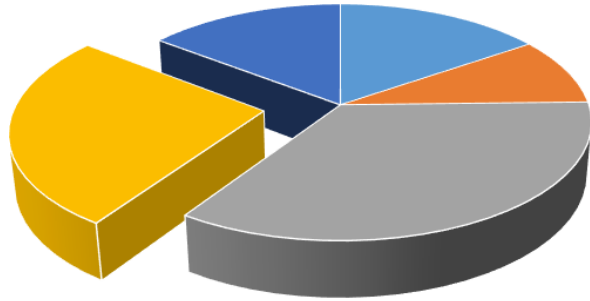


Why Study Optimization?

Resource Allocation

Suppose an online platform must decide **which advertiser's ad to show to each user**.

Target: maximize the total revenue generated over all users



$$\max_{j_1, j_2, \dots, j_T} \sum_{t=1}^T v(i_t, j_t)$$

s.t. advertiser's available **budget**

$$s.t. \sum_{t=1}^T \mathbb{I}\{j_t = j\} \leq G_j \quad \forall j$$

Advertisers
(j)



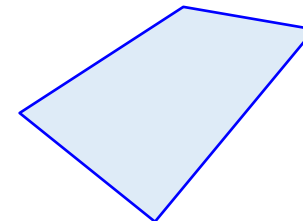
amazon



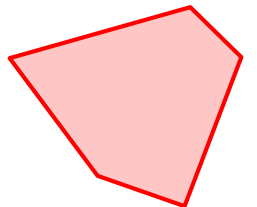
Users
(i)



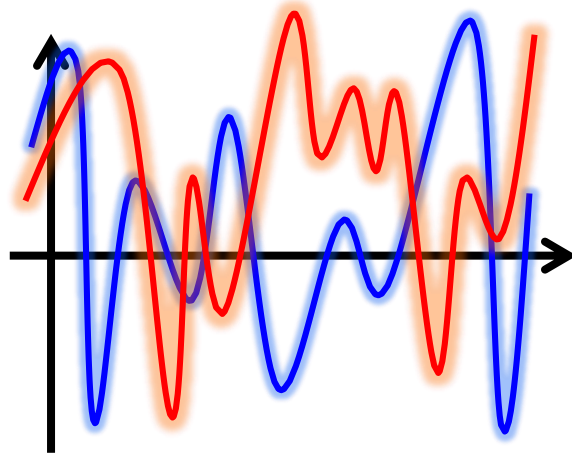
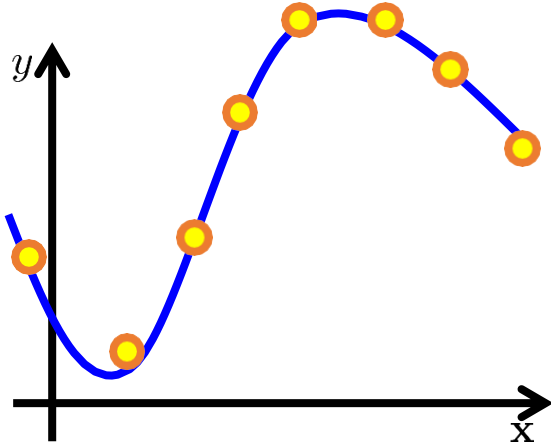
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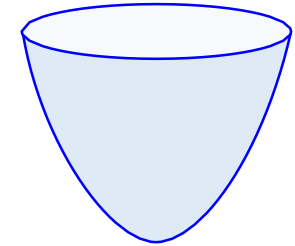
Finance and Economics



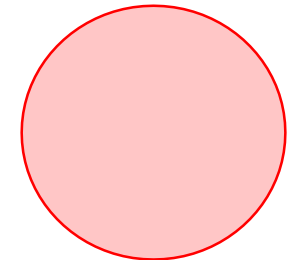
$$\min_{\mathbf{w} \in \mathbb{R}^p} \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \mathbf{w})^2$$
$$s.t. \quad \|\mathbf{w}\|_2 \leq R$$

- Risk analysis in investment profiles
- Consumption assessment
- Demand and Supply prediction
- Trend analysis in time series

f



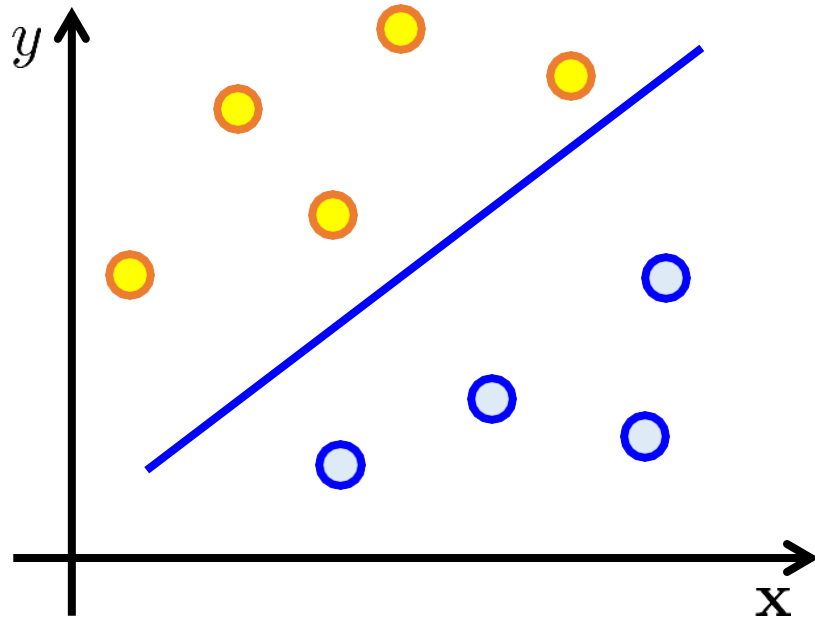
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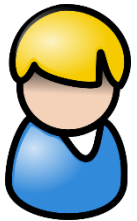
Fraud Detection



Classification problem of Fraud vs Non-Fraud based on user transaction

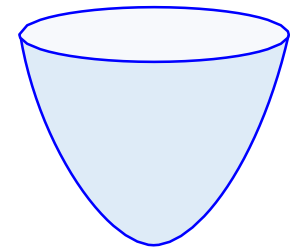


Users



$$\min_{f \in \mathcal{F}} \sum_{i=1}^n \ell(y_i, f(\mathbf{x}_i))$$

f



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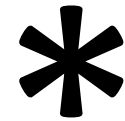


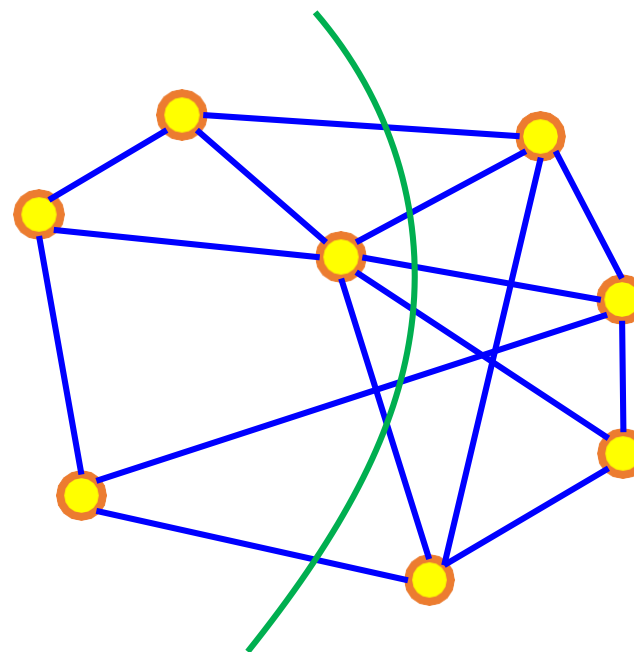
Image Segmentation

Image segmentation can be viewed as finding the best place to cut a graph into S and $V \setminus S$ so that different objects are separated while similar pixels stay together.

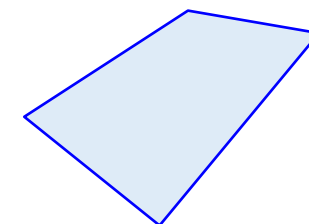
Input



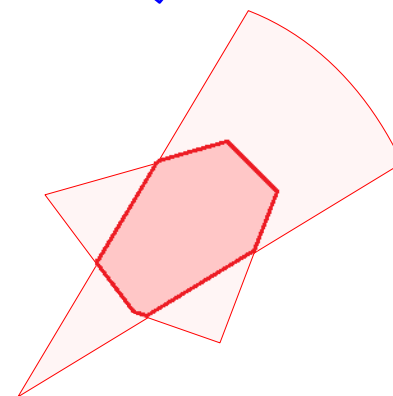
Output



f



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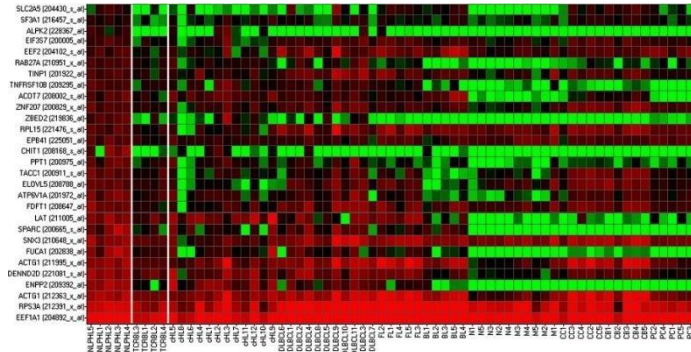
$$\min_{S \subset V} \frac{\text{cut}(S, \bar{S})}{\min\{|S|, |\bar{S}|\}}$$

$$\mathcal{C} = \{S : S \subset V, S \neq \emptyset, S \neq V\}.$$



Why Non-Convex Optimization?

Gene Expression Analysis

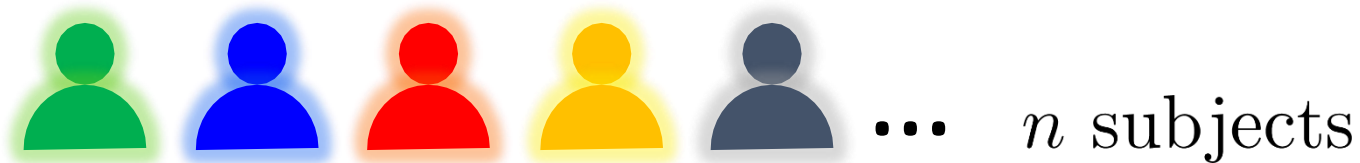


DNA micro-array gene expression data

$$\min_{\mathbf{w} \in \mathbb{R}^p} \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \mathbf{w})^2$$

$$s.t. \quad \|\mathbf{w}\|_0 \leq s \ll p$$

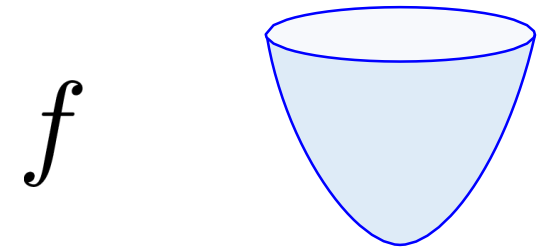
Since small number of genes are truly relevant



$$\text{person}_i = (\mathbf{x}_i, y_i) \in \mathbb{R}^p \times \mathbb{R}$$

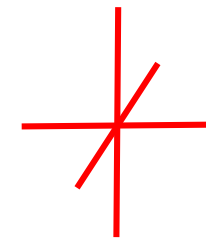
Linear model: $y_i \approx \mathbf{x}_i^\top \mathbf{w}^*$

Challenge: $\mathbf{w}^*$ is sparse!

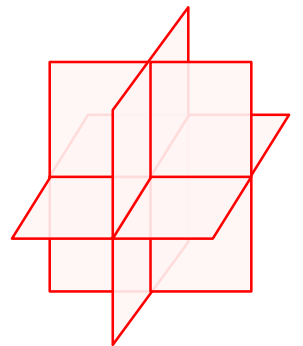


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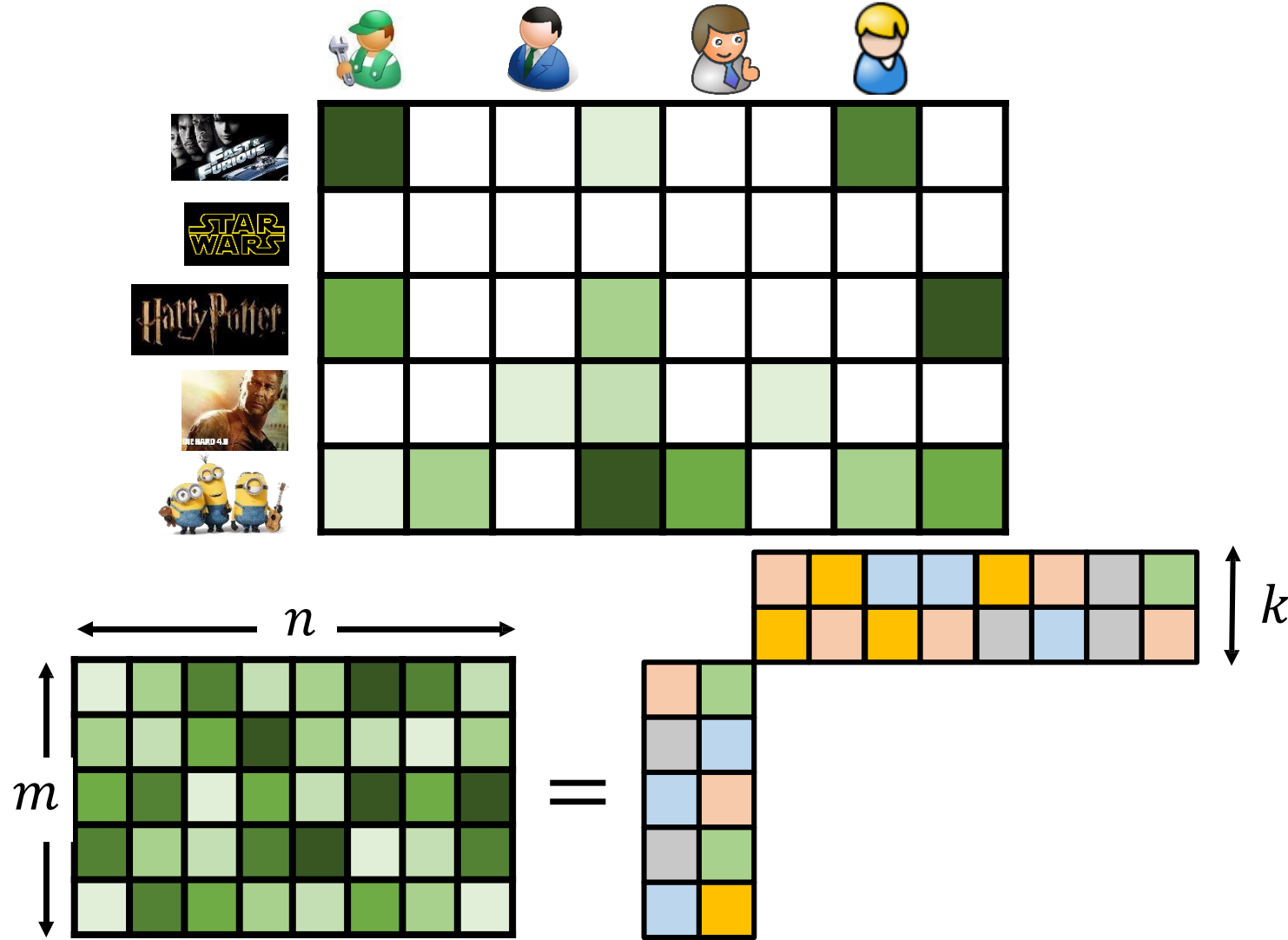


$p = 3, s = 1$



$p = 3, s = 2$

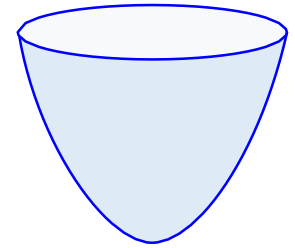
Recommender System: Low-Rank Matrix Completion



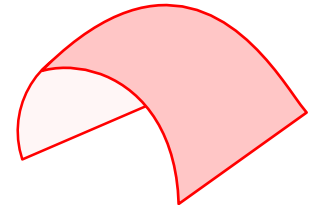
$$\min_{L \in \mathbb{R}^{m \times n}} \|X_{\Omega} - L_{\Omega}\|_F^2$$

$$s.t. \text{rank}(L) \leq k$$

f

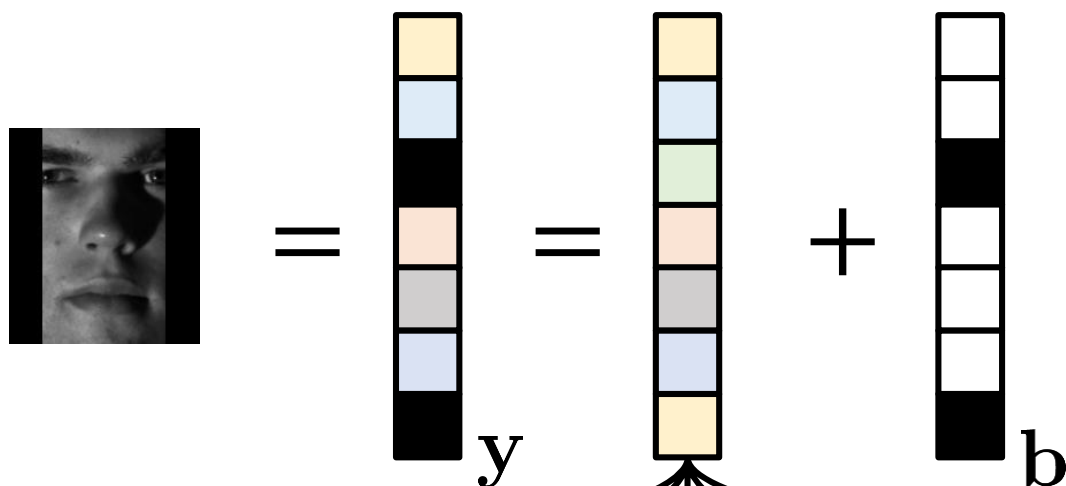


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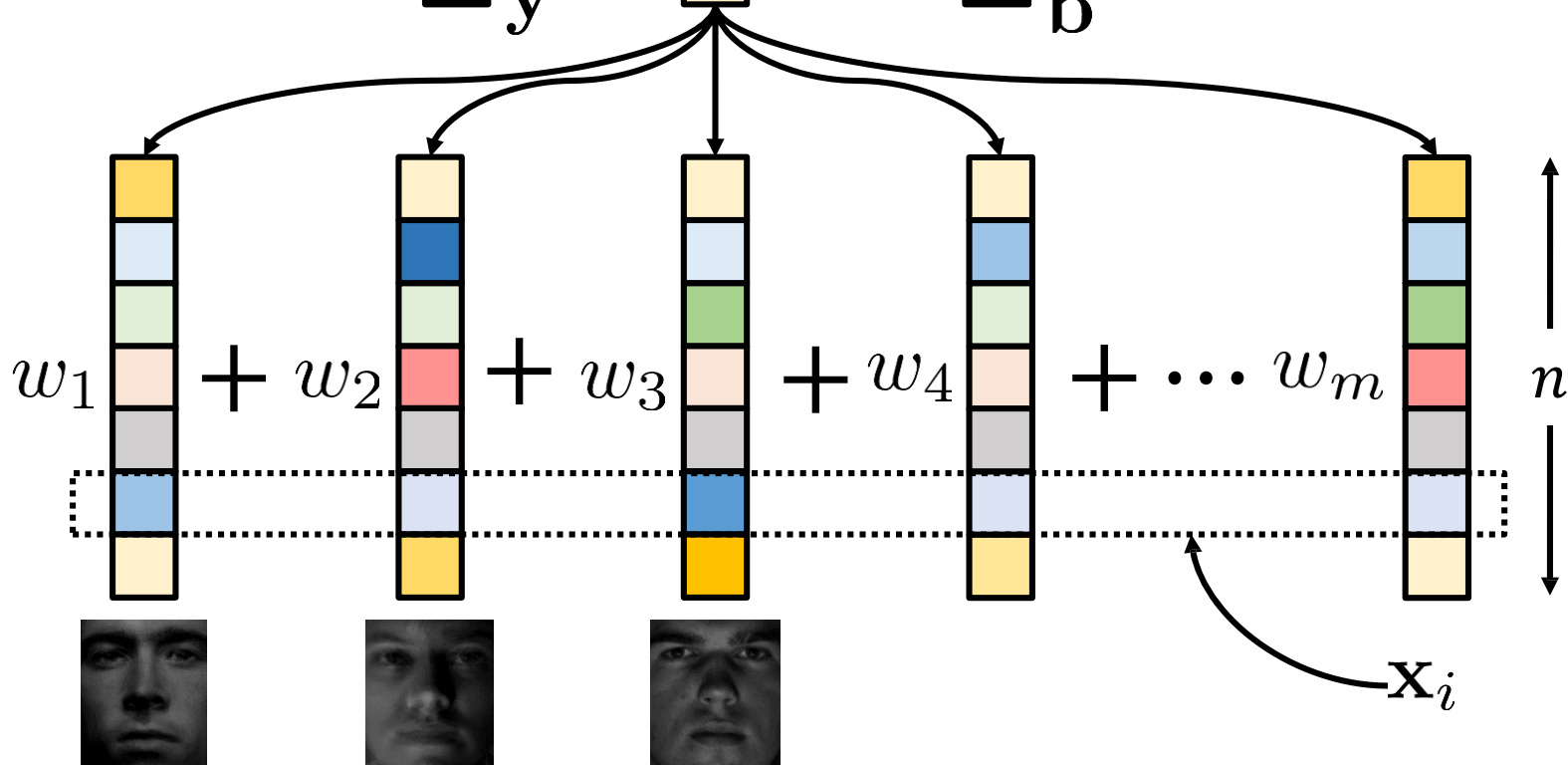
Recommend unseen items by completing a partially observed rating matrix using a low-dimensional latent structure.

Image Denoising and Robust Face Recognition

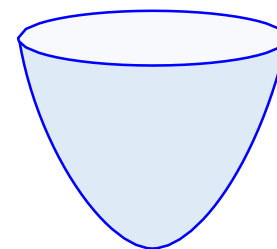


$$\min_{\mathbf{w} \in \mathbb{R}^p} \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \mathbf{w} - b_i)^2$$

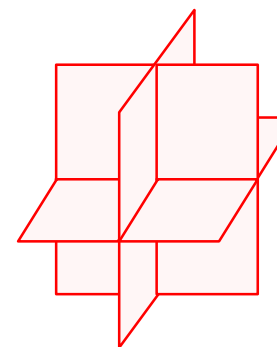
$$s.t. \quad \|\mathbf{b}\|_0 \leq k \ll n$$



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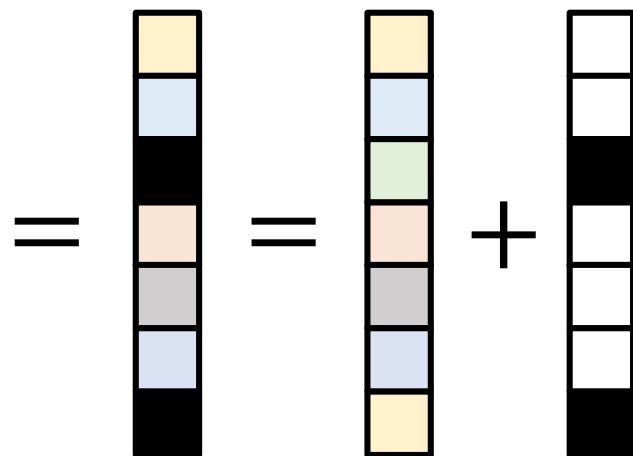


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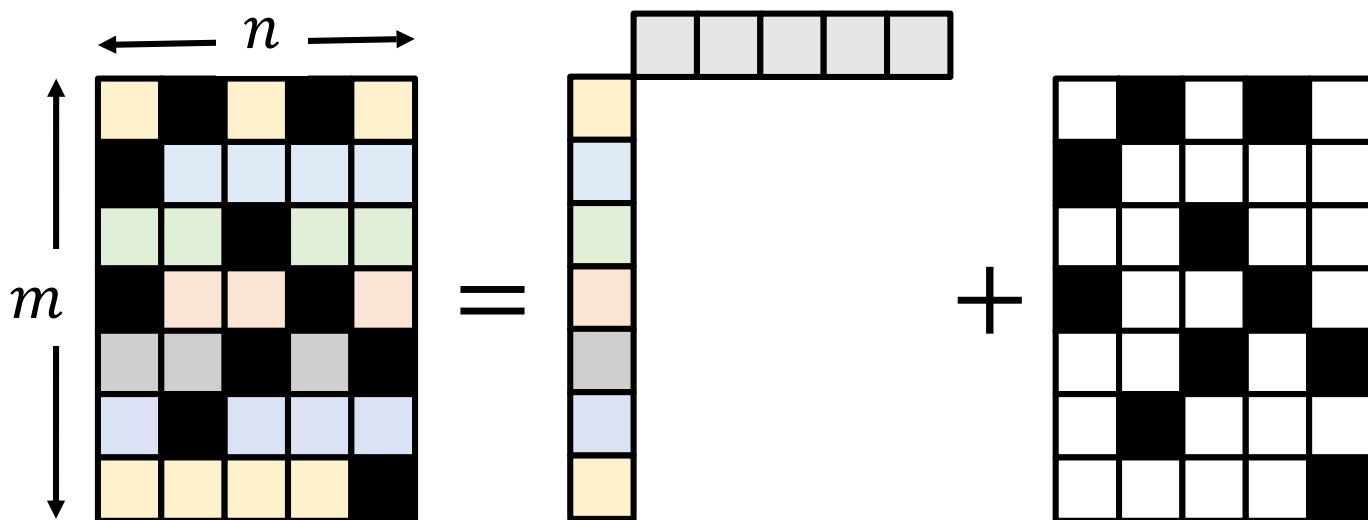
Large Scale Surveillance

- Foreground-background separation

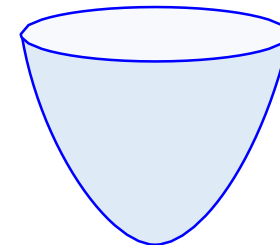


$$\min_{L \in \mathcal{M}_k^{m,n}} \|X - (L + S)\|_F^2$$

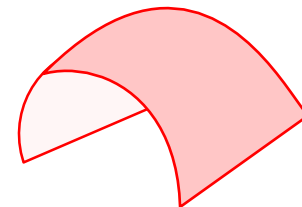
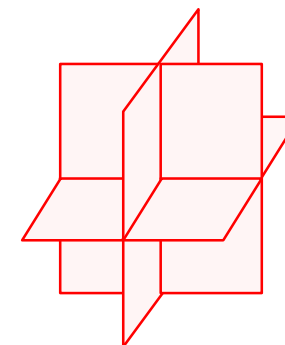
$$s.t. \|S\|_0 \leq s \ll mn$$



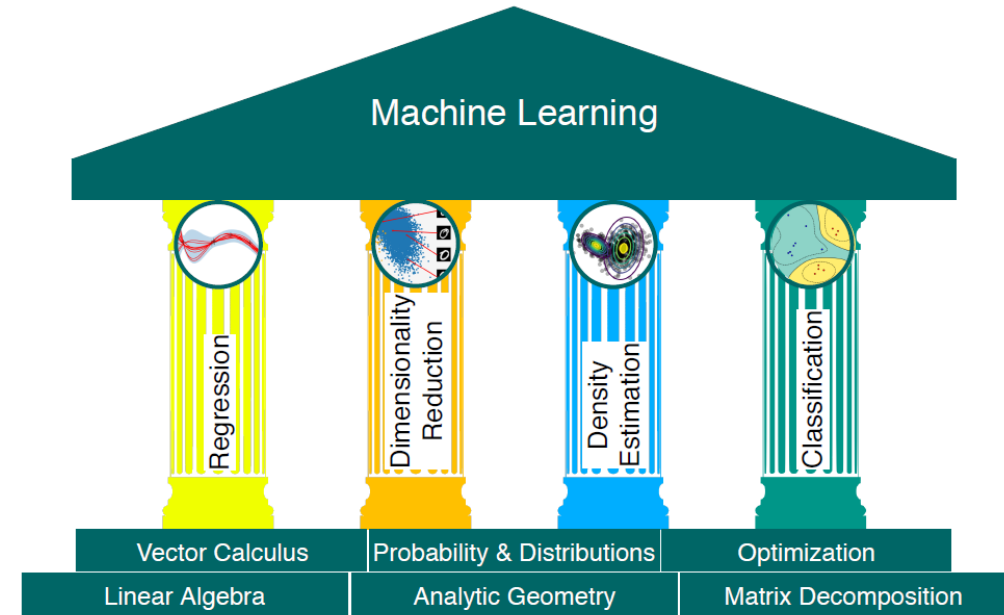
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What we will do in this course



- Vector Calculus
- Optimization Basics
- Introduction to Convexity
- First Order Methods (Gradient descent, Proximal GD, Stochastic GD)
- Second Order Methods (Newton's method and Quasi-Newton methods)
- Constrained Optimization and Lagrange Multipliers
- Non-convex Optimization Methods

Python Toolbox

CVXPY: <https://www.cvxpy.org/>

PYOMO: <http://www.pyomo.org/>

CVXOPT: <https://cvxopt.org/index.html>

Solvers

MOSEK: <https://www.mosek.com/>

Gurobi: <https://www.gurobi.com/>

IPOPT: <https://github.com/coin-or/Ipopt>

COIN-OR: <https://github.com/coin-or/>



Basic Definitions

Optimization for ML

- Regularized Linear Regression (a.k.a. Ridge Regression)

$$\mathbf{w}_{ridge} = \arg \min_{\mathbf{w}} \sum_{n=1}^N (y_n - \mathbf{w}^\top \mathbf{x}_n)^2 + \lambda \mathbf{w}^\top \mathbf{w} = (\sum_{n=1}^N \mathbf{x}_n \mathbf{x}_n^\top + \lambda I_D)^{-1} (\sum_{n=1}^N y_n \mathbf{x}_n)$$

Problem more compactly written as $\arg \min_{\mathbf{w}} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \lambda \|\mathbf{w}\|_2^2$

Solution more compactly as $(\mathbf{X}^\top \mathbf{X} + \lambda I_D)^{-1} \mathbf{X}^\top \mathbf{y}$

- Getting closed-form sol. required simple calculus, but is expensive to compute
 - Especially when D is very large (since we need to invert a $D \times D$ matrix)
- How to solve this and other (possibly more difficult) optimization problems arising in ML efficiently?
- What's the basic calculus and optimization knowledge we need?

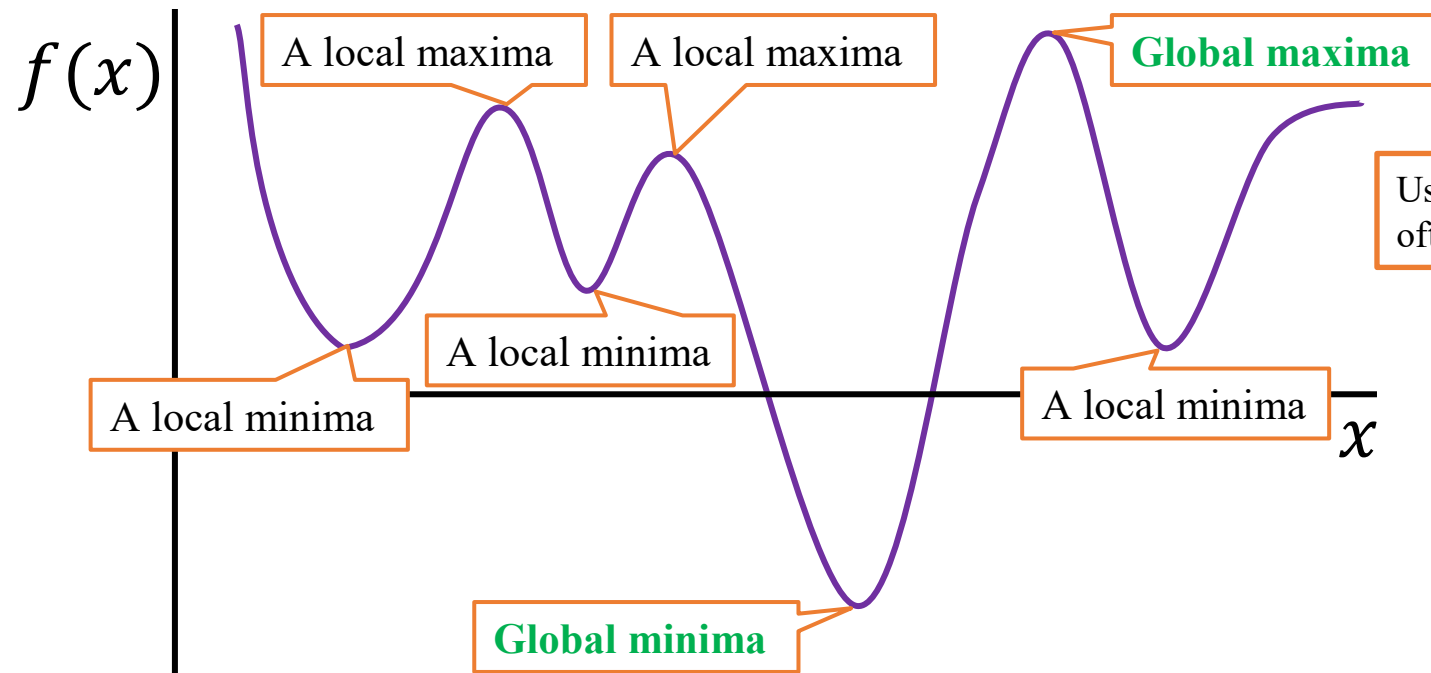
Functions and their optima



- Many ML problems require us to optimize a function f of some variable(s) x
- For simplicity, assume f is a scalar-valued function of a scalar x ($f: \mathbb{R} \rightarrow \mathbb{R}$)

The objective function of the ML problem we are solving (e.g., squared loss for regression)

Assume unconstrained for now, i.e., just a real-valued number/vector



Usually interested in global optima but often want to find local optima, too

For deep learning models, often the local optima are what we can find (and they usually suffice) – more later

Will see what these are later

- Any function has one/more optima (maxima, minima), and maybe saddle points
- Finding the optima or saddles requires derivatives/gradients of the function



Derivatives

Will sometimes use $f'(x)$ to denote the derivative

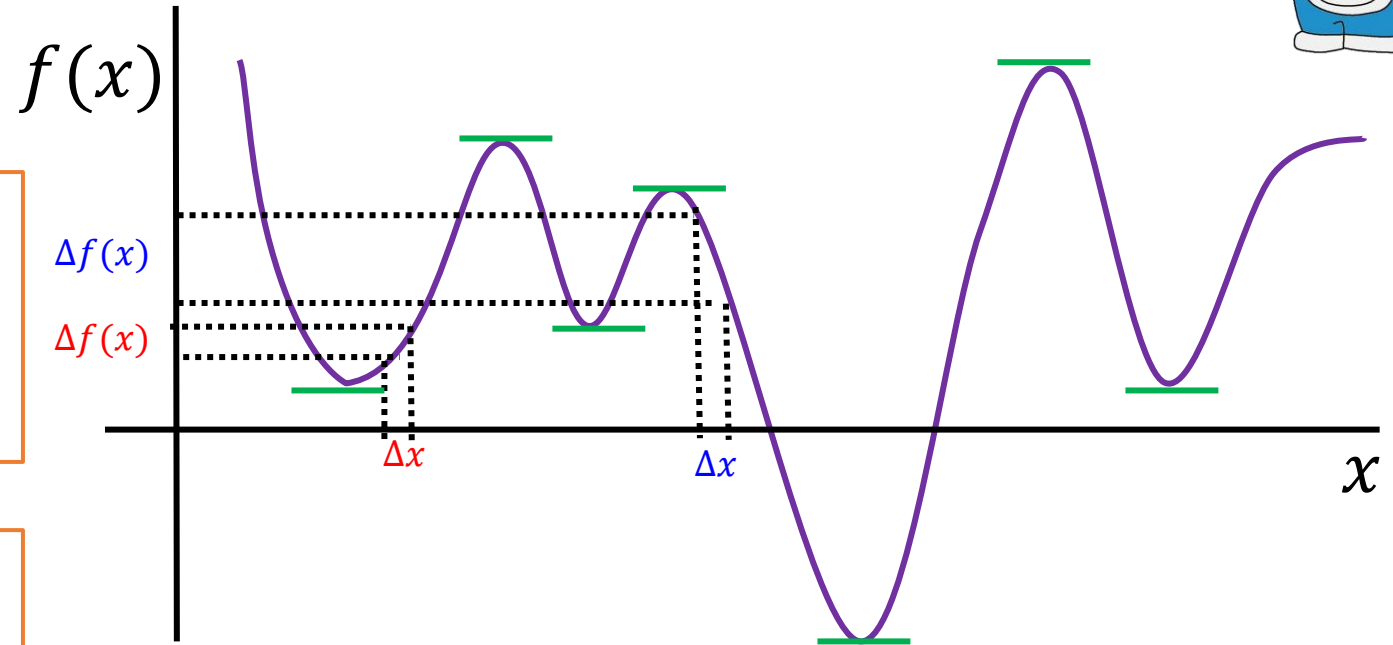


- Magnitude of derivative at a point is the rate of change of the function at that point

$$\frac{df(x)}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta f(x)}{\Delta x}$$

Sign is also important: Positive derivative means f is **increasing** at x if we increase the value of x by a very small amount; negative derivative means it is **decreasing**

Understanding how f changes its value as we change x is helpful to understand optimization (minimization/maximization) algorithms



- Derivative becomes zero at stationary points (optima or saddle points)
 - The function becomes “flat” ($\Delta f(x) = 0$ if we change x by a very little at such points)
 - These are the points where the function has its maxima/minima (unless they are saddles)

Rules of Derivatives

Some basic rules of taking derivatives

- **Sum Rule:** $(f(x) + g(x))' = f'(x) + g'(x)$
- **Scaling Rule:** $(a \cdot f(x))' = a \cdot f'(x)$ if a is not a function of x
- **Product Rule:** $(f(x) \cdot g(x))' = f'(x) \cdot g(x) + g'(x) \cdot f(x)$
- **Quotient Rule:** $(f(x)/g(x))' = (f'(x) \cdot g(x) - g'(x)f(x))/(g(x))^2$
- **Chain Rule:** $(f(g(x)))' \stackrel{\text{def}}{=} (f \circ g)'(x) = f'(g(x)) \cdot g'(x)$

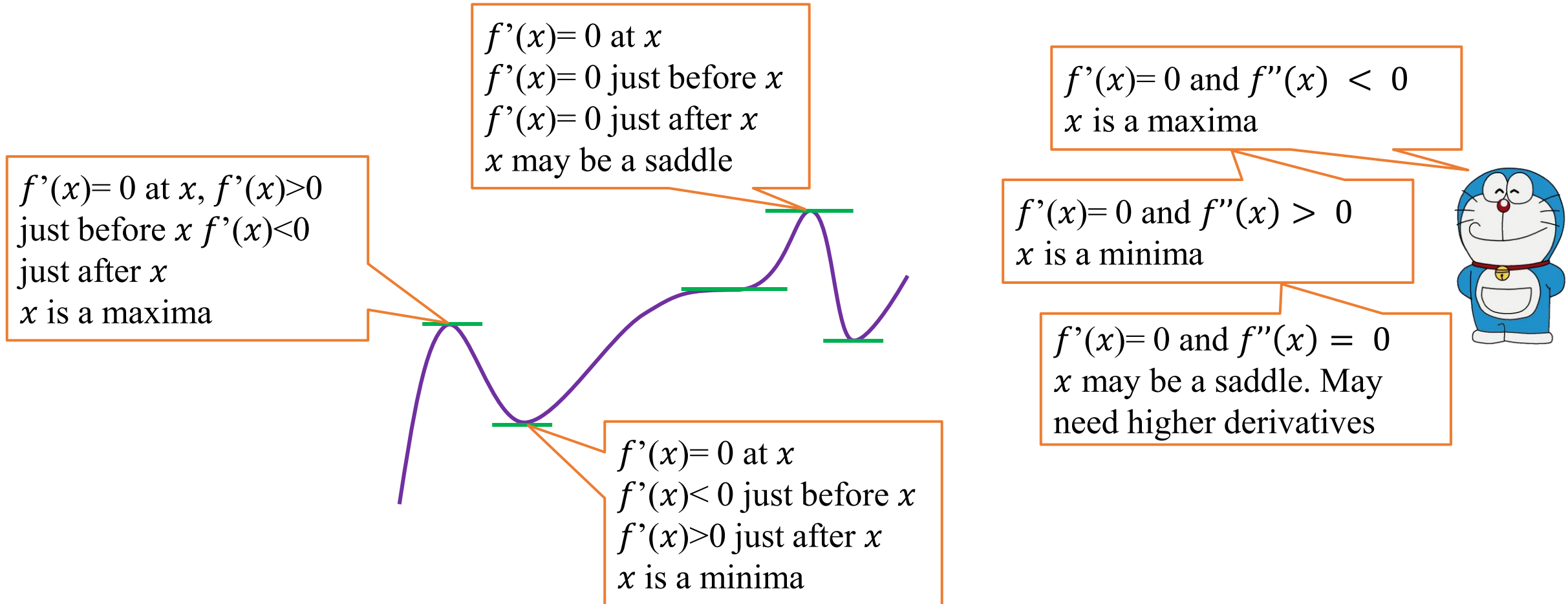


We already used some of these (sum, scaling and chain) when calculating the derivative for the linear regression model

Derivatives



- How the derivative itself changes tells us about the function's optima

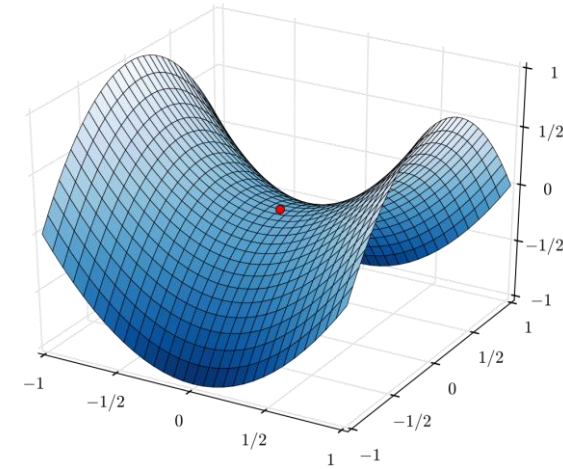
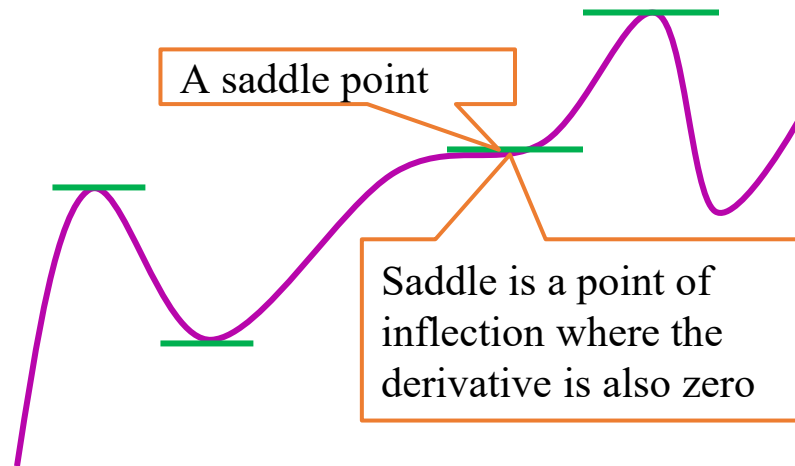


- The second derivative $f''(x)$ can provide this information

Saddle Points



- Points where derivative is zero but are neither minima nor maxima



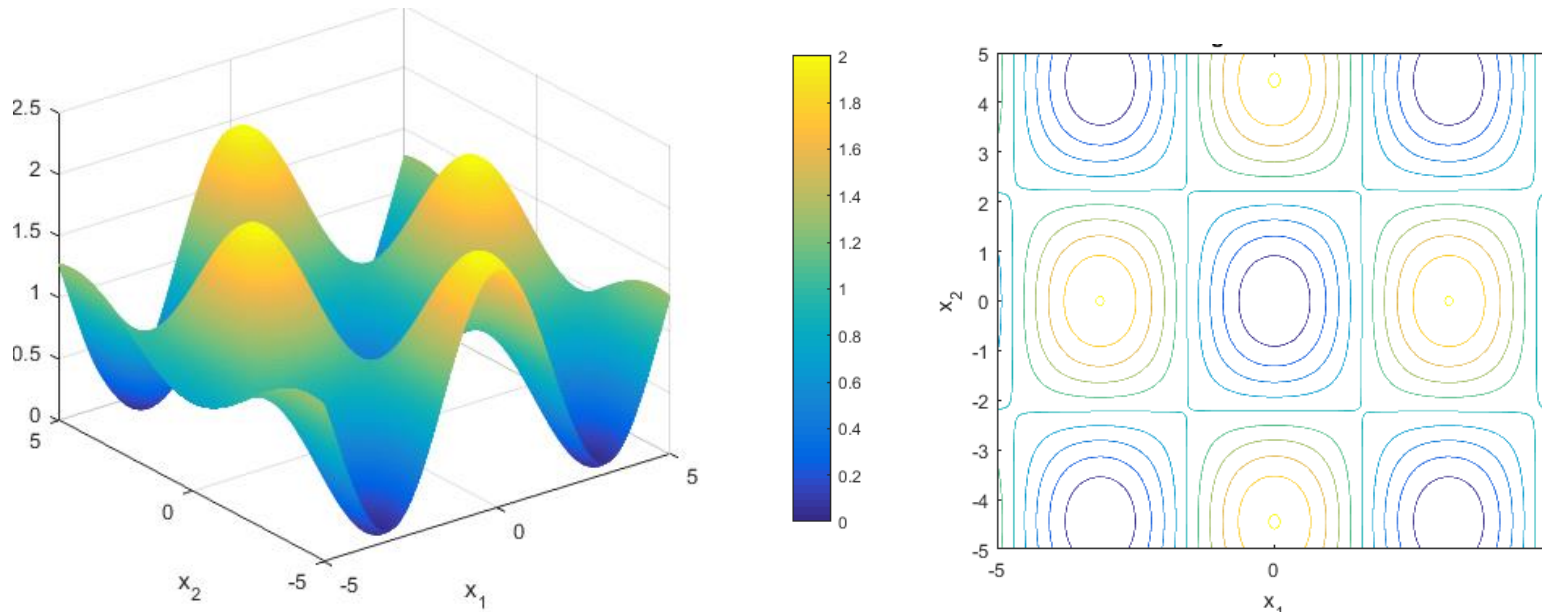
- Saddle points are very common for loss functions of deep learning models
 - Need to be handled carefully during optimization
- Second or higher derivative may help identify if a stationary point is a saddle

Multivariate Functions

- Most functions that we see in ML are multivariate function
- Example: Loss function $L(\mathbf{w})$ in linear regression was a multivariate function of D -dim vector $\mathbf{w}$

$$L(\mathbf{w}): \mathbb{R}^D \rightarrow \mathbb{R}$$

- Here is an illustration of a function of 2 variables (4 maxima and 5 minima)



Two-dim contour plot of the function (i.e., what it looks like from the above)

Derivatives of Multivariate Functions

- Can define derivative for a multivariate functions as well via the gradient
- Gradient of a function $f(\mathbf{x}): \mathbb{R}^D \rightarrow \mathbb{R}$ is a $D \times 1$ vector of partial derivatives

$$\nabla f(\mathbf{x}) = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_D} \right)^T$$

Each element in this gradient vector tells us how much f will change if we move a little along the corresponding (akin to one-dim case)

- Optima and saddle points defined similar to one-dim case
 - Required properties that we saw for one-dim case must be satisfied along all the directions
- The second derivative in this case is known as the **Hessian**

The Hessian



- For a multivar scalar valued function $f(\mathbf{x}): \mathbb{R}^D \rightarrow \mathbb{R}$, Hessian is a $D \times D$ matrix

$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_D} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \dots & \frac{\partial^2 f}{\partial x_2 \partial x_D} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_D \partial x_1} & \frac{\partial^2 f}{\partial x_D \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_D^2} \end{bmatrix}$$

Gives information about the **curvature** of the function at point $\mathbf{x}$

Note: If the function itself is vector valued, e.g., $f(\mathbf{x}): \mathbb{R}^D \rightarrow \mathbb{R}^K$ then we will have K such $D \times D$ Hessian matrices, one for each output dimension of f

A square, symmetric $D \times D$ matrix $\mathbf{M}$ is PSD if $\mathbf{x}^\top \mathbf{M} \mathbf{x} \geq \mathbf{0} \forall \mathbf{x} \in \mathbb{R}^D$
Will be NSD if $\mathbf{x}^\top \mathbf{M} \mathbf{x} \leq \mathbf{0} \forall \mathbf{x} \in \mathbb{R}^D$

PSD if all eigenvalues are non-negative

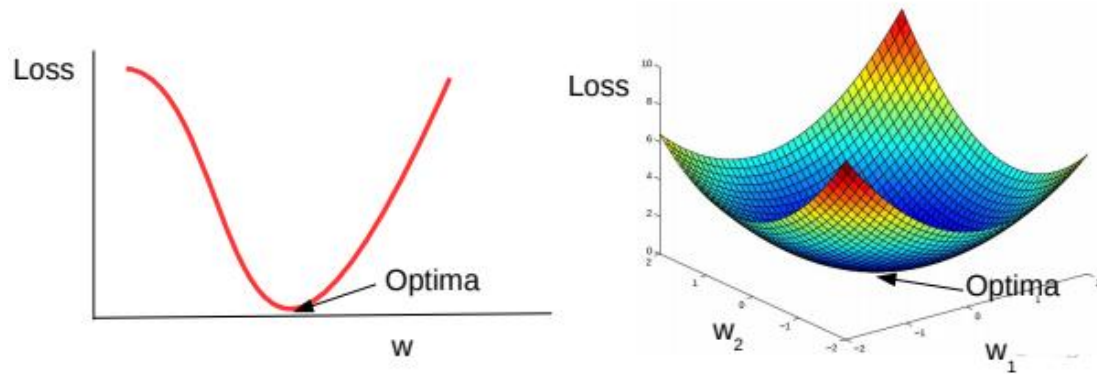
- The Hessian matrix can be used to assess the optima/saddle points
 - $\nabla f(\mathbf{x}) = 0$ and $\nabla^2 f(\mathbf{x})$ is a positive definite (PD) matrix then $\mathbf{x}$ is a minima
 - $\nabla f(\mathbf{x}) = 0$, and $\nabla^2 f(\mathbf{x})$ is a negative definite (ND) matrix then $\mathbf{x}$ is a maxima



Convex and Non-Convex Functions



- A function being optimized can be either **convex** or **non-convex**
- Here are a couple of examples of convex functions

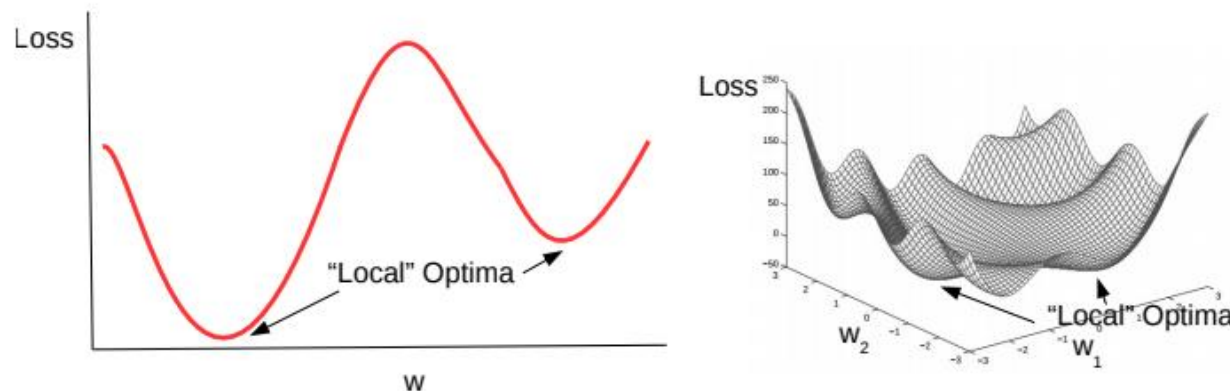


Convex functions are bowl-shaped.
They have a unique optima (minima)

Negative of a convex function is
called a **concave** function, which also
has a unique optima (maxima)

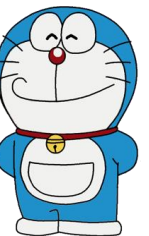


- Here are a couple of examples of non-convex functions



Non-convex functions have
multiple minima. Usually
harder to optimize as
compared to convex functions

Loss functions of most
deep learning models are
non-convex



Convex Sets



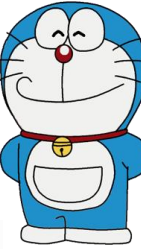
- A set S of points is a convex set, if for any two points $x, y \in S$, and $0 \leq \alpha \leq 1$

z is also called a
“convex combination”
of two points

$$z = \alpha x + (1 - \alpha)y \in S$$

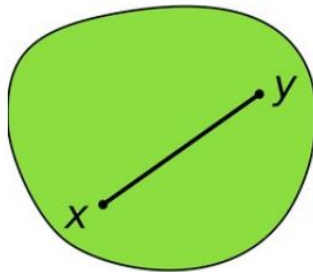
Can also define convex
combination of N
points $x_1, x_2, \dots, x_N$ as

$$z = \sum_{i=1}^N \alpha_i x_i, \quad \alpha_i \geq 0, \quad \sum_{i=1}^N \alpha_i = 1.$$

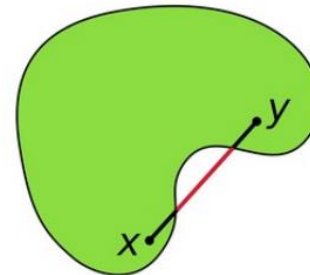


- Above means that all points on the line-segment between x and y lie within S

A Convex Set



A Non-convex Set

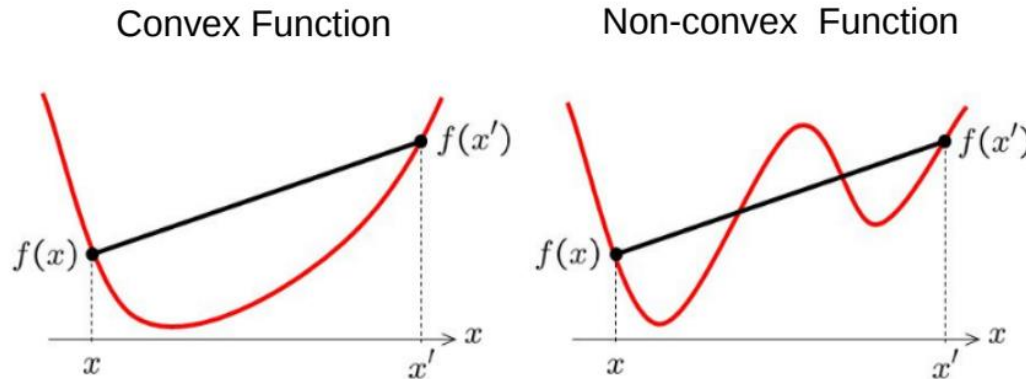


- The domain of a convex function needs to be a convex set

Convex Functions



- Informally, $f(x)$ is convex if for any two points on the graph, the line segment joining them lies on or above the graph.

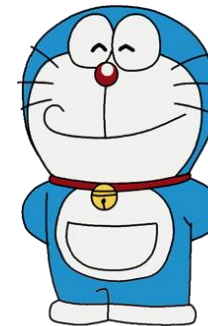
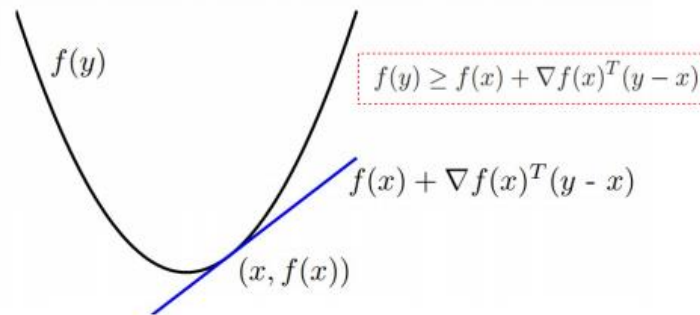


Note: "Chord lies above function"
more formally means

If f is convex then given
 $\alpha_1, \dots, \alpha_n$ s.t. $\sum_{i=1}^n \alpha_i = 1$
$$f\left(\sum_{i=1}^n \alpha_i x_i\right) \leq \sum_{i=1}^n \alpha_i f(x_i)$$

Jensen's Inequality

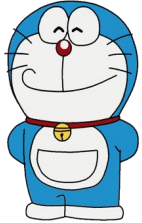
- Formally, (assuming differentiable function), some tests for convexity:
 - First-order convexity (graph of f must be above all the tangents)



Exercise: Show that
ridge regression
objective is convex

- Second derivative a.k.a. Hessian (if exists) must be positive semi-definite

Convex Functions: Simple Proof



Exercise: Show that ridge regression objective is convex.

$$f(w) = \|y - Xw\|_2^2 + \lambda \|w\|_2^2$$

$$\nabla f(w) = -2X^\top y + 2X^\top Xw + 2\lambda w$$

$$\nabla^2 f(w) = 2X^\top X + 2\lambda I$$

Since

$$X^\top X \succeq 0, \quad \lambda I \succeq 0,$$

we have

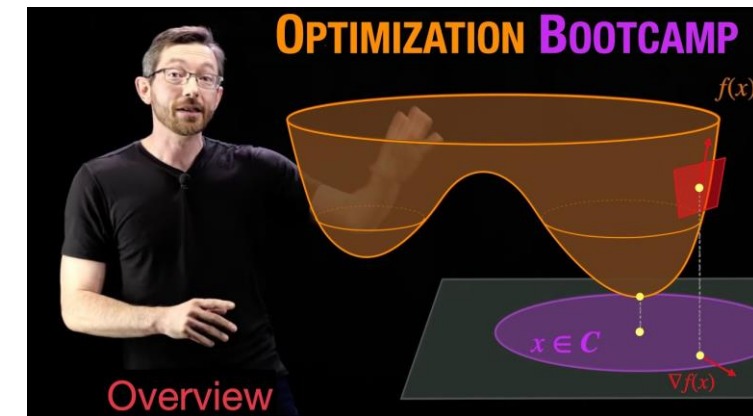
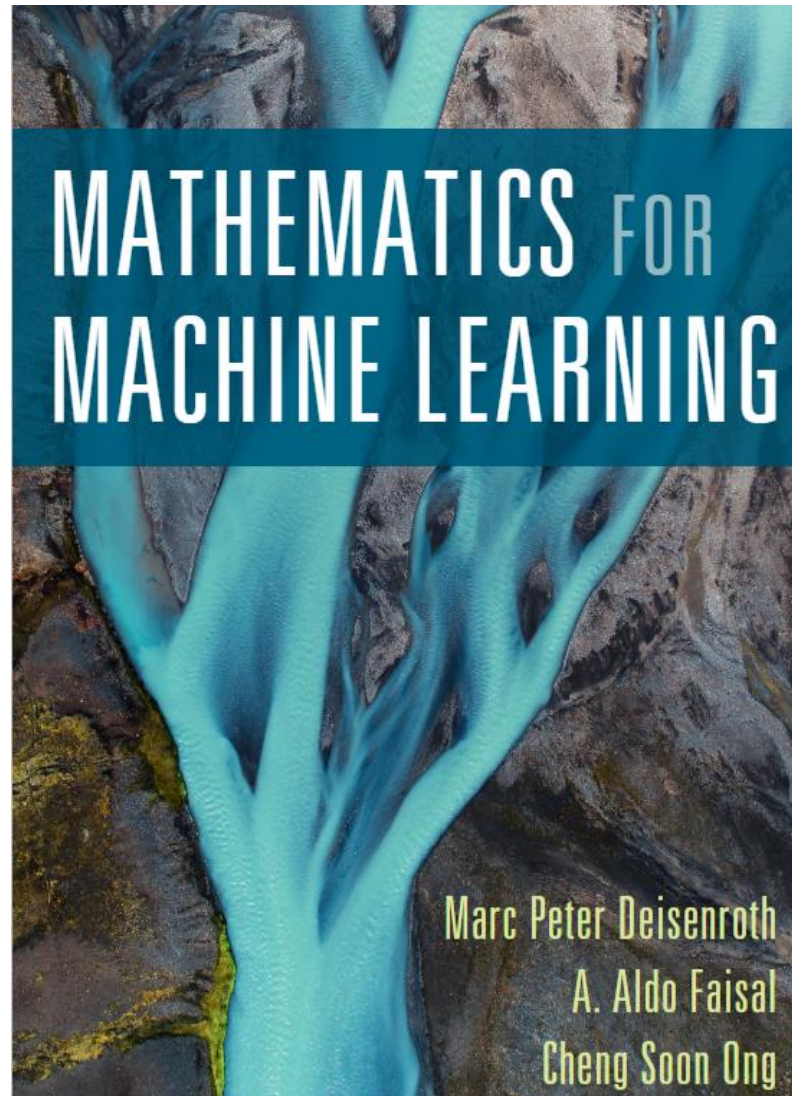
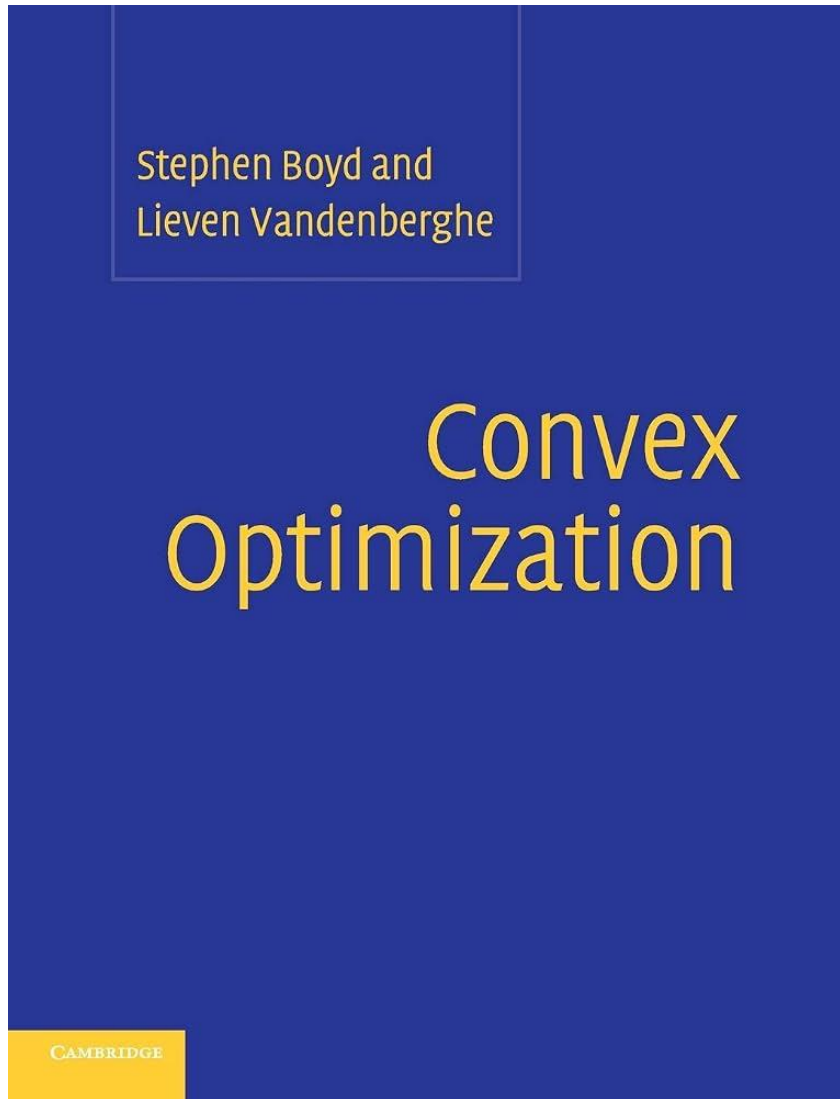
$$\nabla^2 f(w) \succeq 0$$

and therefore

the ridge regression objective is convex.

For $\lambda > 0$, it is strictly convex.

References



Links:

- [MML Book](#)
- [Convex Optimization Book](#)
- [Optimization Bootcamp](#)

Slide Credits: Some slides have been adapted from openly available course materials on Optimization by [Puru](#) and Machine Learning by [Piyush](#).